Week 1-2: Python Basics (Days 1-10)

Learning Objectives

- Understand Python syntax and basic programming concepts
- Write functional Python programs
- Build a strong foundation for data analysis

Detailed Topics

Days 1-3: Fundamentals

- Variables & Types: int, float, string, bool; type conversion, naming rules
- Operators: Arithmetic (+, -, *, /, %, **); Comparison (==, !=, >, <, >=, <=); Logical (and, or, not)

Days 4-6: Control Flow

- Conditionals: if/elif/else, nested conditions, ternary operators
- Loops: for (range, enumerate), while; break, continue, pass; loop else

Days 7-8: Data Structures

- Lists: indexing/slicing, methods (append, extend, insert, remove, pop, sort), comprehensions
- Tuples & Sets: tuple creation/unpacking; set operations (union, intersection, difference)
- Dictionaries: creation/access, methods, comprehensions

Days 9-10: Functions & Programs

- Functions: definition, parameters/arguments, return, default/keyword args, *args/**kwargs, lambda
- Basic Programs: calculator, number guessing game, data processor, file I/O
- Projects: calculator app, to-do list manager, text analyzer

Practice Projects

1. Build a simple calculator
2. Create a to-do list manager
3. Develop a basic text analyzer